

A one-parameter family of critical point inequalities

(on Problem 29 of your open problems list)

Claude (Fable 5, Anthropic)*

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Problem 29 of your list concerns the inequality (*Wasserstein Distance, Fourier Analysis...*, 2018): for $f : \mathbb{T} \rightarrow \mathbb{R}$ continuously differentiable with mean value 0,

$$N(f) \cdot \|f\|_{L^2(\mathbb{T})} \gtrsim \frac{\|f'\|_{L^1(\mathbb{T})}^2}{\|f'\|_{L^\infty(\mathbb{T})}}, \quad (1)$$

where $N(f)$ denotes the number of critical points, together with the elementary companion $N(f) \cdot \|f\|_{L^\infty} \gtrsim \|f'\|_{L^1}$. You ask whether there are more such inequalities and whether they are part of a family. They are: the following one-parameter family contains the elementary inequality as the endpoint $p = \infty$ and *strengthens* (1) at $p = 2$. Throughout, $\mathbb{T}$ is normalized to have total measure 1.

Theorem 1. *Let $f : \mathbb{T} \rightarrow \mathbb{R}$ be C^1 , nonconstant, with finitely many critical points $N = N(f)$. Then for every $1 \leq p < \infty$,*

$$N(f) \cdot \|f\|_{L^p(\mathbb{T})} \geq \frac{1}{4} \cdot 2^{-1/p} \frac{\|f'\|_{L^1(\mathbb{T})}^{1+1/p}}{\|f'\|_{L^\infty(\mathbb{T})}^{1/p}}. \quad (2)$$

Note that the mean value zero assumption is not needed.

Proof. Write $L = \|f'\|_{L^\infty}$ and $V = \|f'\|_{L^1}$. Since f is nonconstant and periodic, f' vanishes somewhere, so $N \geq 2$ (both extrema are critical). The critical points partition $\mathbb{T}$ into N closed arcs $I_1, \dots, I_N$ on whose interiors f' is continuous and nonvanishing; hence f is strictly monotone on each arc. Let

$$v_j = \int_{I_j} |f'| dx = |f(b_j) - f(a_j)|, \quad \sum_{j=1}^N v_j = V.$$

Step 1: each arc carries mass. Fix j and abbreviate $v = v_j$. The monotone map f sends I_j bijectively onto a value interval $J_j \subset \mathbb{R}$ of length v . The set $\{y \in J_j : |y| \leq v/4\}$ is contained in an interval of length $v/2$, so

$$|\{y \in J_j : |y| \geq v/4\}| \geq v/2.$$

*This note was written by an AI language model in conversation with Ethan Kuntz (mindthemanifold.substack.com), who suggested the problem list and independently checked the argument; correspondence can be directed to him. Numerical stress-testing (random trigonometric polynomials and the spike family below, $p \in \{1, 2, 4\}$) found no violation of (2) and located near-extremizers at narrow spikes, consistent with the proof mechanism.

Since f is Lipschitz with constant L , images of sets shrink by at most a factor L ; equivalently, for any measurable $S \subseteq J_j$,

$$|f^{-1}(S) \cap I_j| \geq \frac{|S|}{L}.$$

Applying this with $S = \{y \in J_j : |y| \geq v/4\}$,

$$\int_{I_j} |f|^p dx \geq \left(\frac{v}{4}\right)^p \cdot \frac{v}{2L} = \frac{v^{p+1}}{2 \cdot 4^p L}.$$

Step 2: convexity. Summing over the arcs and using Jensen's inequality for the convex function $t \mapsto t^{p+1}$,

$$\|f\|_{L^p}^p \geq \frac{1}{2 \cdot 4^p L} \sum_{j=1}^N v_j^{p+1} \geq \frac{1}{2 \cdot 4^p L} \cdot N \left(\frac{V}{N}\right)^{p+1} = \frac{V^{p+1}}{2 \cdot 4^p L N^p}.$$

Taking p -th roots and rearranging yields (2). $\square$

Remark 1 (The family contains both known inequalities). *Letting $p \rightarrow \infty$ in (2) recovers $N\|f\|_{L^\infty} \geq \frac{1}{4}\|f'\|_{L^1}$. At $p = 2$, since $|\mathbb{T}| = 1$ gives $\|f'\|_{L^1} \leq \|f'\|_{L^\infty}$, we have*

$$\frac{\|f'\|_{L^1}^{3/2}}{\|f'\|_{L^\infty}^{1/2}} \geq \frac{\|f'\|_{L^1}^2}{\|f'\|_{L^\infty}},$$

so (2) with $p = 2$ implies (1) (without requiring mean value zero).

Remark 2 (Sharpness). *The exponents in (2) are optimal for every p , and there is an explicit witness requiring no auxiliary construction. For $w > 0$ consider*

$$f_w(x) = \exp(-\sin^2(\pi x)/w^2), \quad x \in \mathbb{T},$$

a smooth spike of height 1 and effective width $\sim w$. Its derivative $f'_w(x) = -\pi w^{-2} \sin(2\pi x) f_w(x)$ vanishes exactly at $x \in \{0, \frac{1}{2}\}$, so $N(f_w) = 2$. Laplace asymptotics as $w \rightarrow 0$ give

$$\|f_w\|_{L^p} \sim \left(\frac{w}{\sqrt{p\pi}}\right)^{1/p}, \quad \|f'_w\|_{L^1} \rightarrow 2, \quad \|f'_w\|_{L^\infty} \sim \frac{\pi\sqrt{2}}{e^{1/2}w},$$

so that

$$\frac{N(f_w) \|f_w\|_{L^p}}{\|f'_w\|_{L^1}^{1+1/p} \|f'_w\|_{L^\infty}^{-1/p}} \rightarrow \left(\frac{\sqrt{2\pi/p}}{2e^{1/2}}\right)^{1/p} > 0,$$

confirmed numerically to four digits (0.7602, 0.7332, 0.7852 for $p = 1, 2, 4$). In particular the optimal constant c_p in (2) satisfies $\frac{1}{4}2^{-1/p} \leq c_p \leq (\sqrt{2\pi/p}/(2e^{1/2}))^{1/p}$, e.g. $0.177 \leq c_2 \leq 0.734$; the lower bound is not optimized (the “quarter of the value range” step is lossy). The spike also shows the exponent $1/p$ on $\|f'\|_{L^\infty}$ cannot be lowered: homogeneity under $f \mapsto \lambda f$ forces any inequality of the form $N\|f\|_{L^p} \gtrsim \|f'\|_{L^1}^{1+\beta} \|f'\|_{L^\infty}^{-\beta}$ to have exponents differing by exactly 1, and letting $w \rightarrow 0$ in f_w forces $\beta \geq 1/p$. The example $f = \sin(2\pi kx)$, for which both sides of (2) grow like k , shows the linear dependence on N is also optimal.

Remark 3 (Higher dimensions). *The proof isolates the mechanism: a monotone stretch of variation v must spend time $\geq v/(2L)$ at height $\geq v/4$, and convexity penalizes distributing the total variation unevenly across the N stretches. In $d \geq 2$ the arc decomposition has no direct analogue, and the natural substitute is the coarea formula: the role of N should be played by a quantity measuring the level set geometry, e.g. $\sup_t \mathcal{H}^{d-1}(\{f = t\})$ or the number of components of level sets. This appears related in spirit to the Wasserstein nodal set inequality of Problem 69 (Cavalletti–Farinelli), which may be the correct d -dimensional avatar of this family.*